Advanced high school · geometry

Rotations and complex numbers

Synthesis problem

Investigate, justify, conclude

Define $z_0 = 2$ and $z_{n+1} = iz_n + 2$. Find the fixed point a and an explicit formula for z_n. Determine the minimal period and the sum $z_0 + z_1 + z_2 + z_3$. Interpret the four points geometrically.